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First-in-Class Topical Therapy for GA

HBPB1-007: Topical Therapeutic Candidate for Geographic Atrophy

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Purpose

Geographic atrophy (GA), advanced dry AMD, causes irreversible loss of photoreceptors and retinal pigment epithelium (RPE).

Approved GA treatments — Syfovre & Izervay — require monthly/bimonthly **intravitreal injections (IVT)**.

Global AMD market: **$27.5B by 2031**; GA drugs account for 39.2% of US AMD market.

52% of GA patients prefer eye drops as their first-line treatment.

> **No topical (eye drop) treatment approved for GA — major unmet clinical need**

Hypothesis: HBPB1-007, optimized for ocular penetration, would:

1. Reach therapeutically relevant levels in the **posterior segment** via topical dosing
2. Slow retinal degeneration in a **GA mouse model** ($NaIO_3$-induced)
3. Demonstrate a **favorable ocular safety profile** with no systemic toxicity

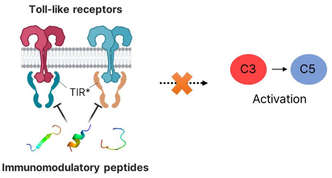
01 Eye drop therapy

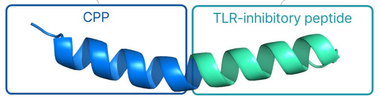
CPP(Cell-penetrating peptide)-based posterior ocular delivery

02 Anti-inflammatory peptide discovery

The biggest TLR-targeted custom peptide library

Phage display-based massively parallel screening of > 190,000 peptides
→ Hundreds of potential TLR inhibitory peptides





HBPB1-007

Composition
32 amino acid peptide

Molecular Weight
3,982.71 Da

Methods — Drug Discovery & Development

1. **T-Surf Peptide Library Construction** *Adv. Sci., 12,* 2406018 (2025)
   – TIR domain family → surface segments → 190,945 peptides → M13 phage
2. **Phage Screening & In Vitro Validation**
   – Target: MAL/MyD88 · Top candidate peptides synthesized with CPP
   – Cell models: HEK-Blue hTLR · RAW264.7 · BMDM · ARPE19
3. **In Vivo Lead Optimization**
   – Efficacy evaluated in a $NaIO_3$-induced mouse GA model via topical instillation
   → **HBPB1-007 (TDIP5)** selected as lead molecule
4. **Pre-Clinical Development**

Results — In Vitro MoA

01 TLR4 Signaling Inhibition

HBPB1-007 binds MAL/MyD88 TIR domains, disrupting TIR-TIR interactions required for TLR4 signalosome assembly.

Assay Platforms

HEK-Blue hTLR4
SEAP reporter assay

RAW264.7
LPS-stimulated macrophage, WB & ELISA

ARPE-19
Human RPE, LPS-stim inflammation

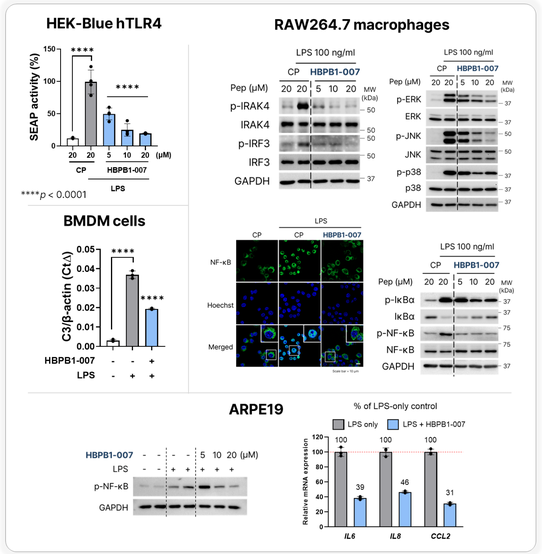
BMDM
C3 mRNA (qRT-PCR)

Cell-type specific readouts

RAW264.7
p-IRAK4, MAPK, p-NF-κB↓ (WB)
IL-6, TNF-α ↓ (ELISA)

ARPE-19
IL-6, IL-8, CCL2 ↓
(RPE paracrine amplification ↓)

BMDM
C3 mRNA ↓
(qRT-PCR, LPS-primed)

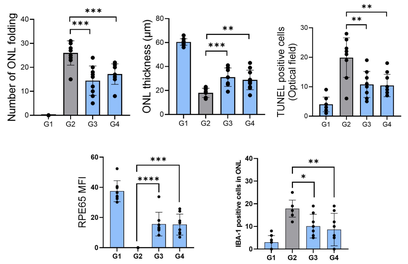


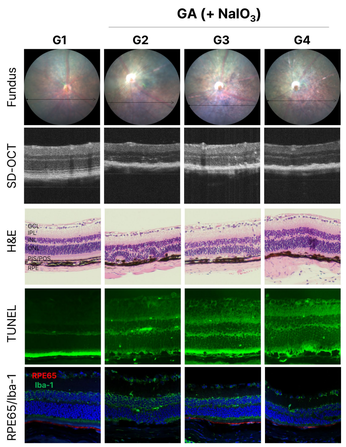
Results — In Vivo Efficacy (GA Model)

02 $NaIO_3$-induced GA mouse model

C57BL/6J mice
(Female, 6 weeks)

Group (n=10/group)	Treatment
G1	Normal (PBS)
G2	Vehicle ($NaIO_3$+PBS)
G3	$NaIO_3$ + HBPB1-007 (20 μM, 0.01%)
G4	$NaIO_3$ + HBPB1-007 (200 μM, 0.1%)

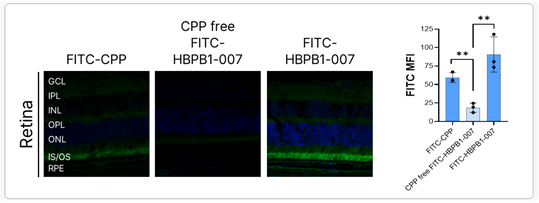


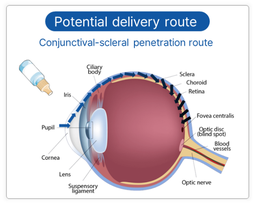


Results — Ocular Drug Delivery

03 Posterior Segment Penetration (FITC Imaging)

C57BL/6J mice
(Female, 6 weeks)





Results — Ocular Safety

04 4-Week Repeat-Dose Preliminary Toxicity Study in NZW Rabbit

Group	Test Item	N (M/F)	Treatment
G1 (Control)	1×PBS	3/3	50 μL × 3 drops (5-min interval) × 2 sessions (6 h apart) → 300 μL/head/day × 28 days (4 weeks)
G2	HBPB1-007 (0.1%)	3/3	
G3	HBPB1-007 (0.5%)	3/3	

No test-article-related toxicological changes.

No adverse findings across clinical signs, body weight, IOP, ophthalmology, hematology, clinical chemistry, organ weights, necropsy, or histopathology.

- ✓ **IOP maintained within normal range (10–21 mmHg) at all timepoints**
- ✓ **NOAEL: 300 μL/head/day (0.5% solution)**

Pre-Clinical Development Plan (w/ Huons Biopharma)

- **2025** ✓ FDA Pre-IND · 4-week rabbit toxicity study
- **2026** DS/DP manufacturing · Formulation · PK
- **2027** GLP-Tox (Rabbit 39w + Dog 26w + Rat SC 26w) · IND preparation
- **2028** IND filing · Phase 2 initiation

Conclusions

1. HBPB1-007 suppresses TLR4 signaling pathways, thereby reducing NF-κB, cytokines, and C3 in RPE cells.
2. Topical dosing significantly preserves ONL thickness and reduces inflammation in a GA mouse model.
3. Non-invasive eye drops effectively deliver the peptide to the posterior segment.
4. Favorable ocular safety profile in a 4-week rabbit preliminary toxicity study (NOAEL: 300 μL/head/day of 0.5% solution).

HBPB1-007 warrants further IND-enabling studies as a patient-friendly, non-invasive eye drop treatment for geographic atrophy.